

# Supervised Learning (Target variable)

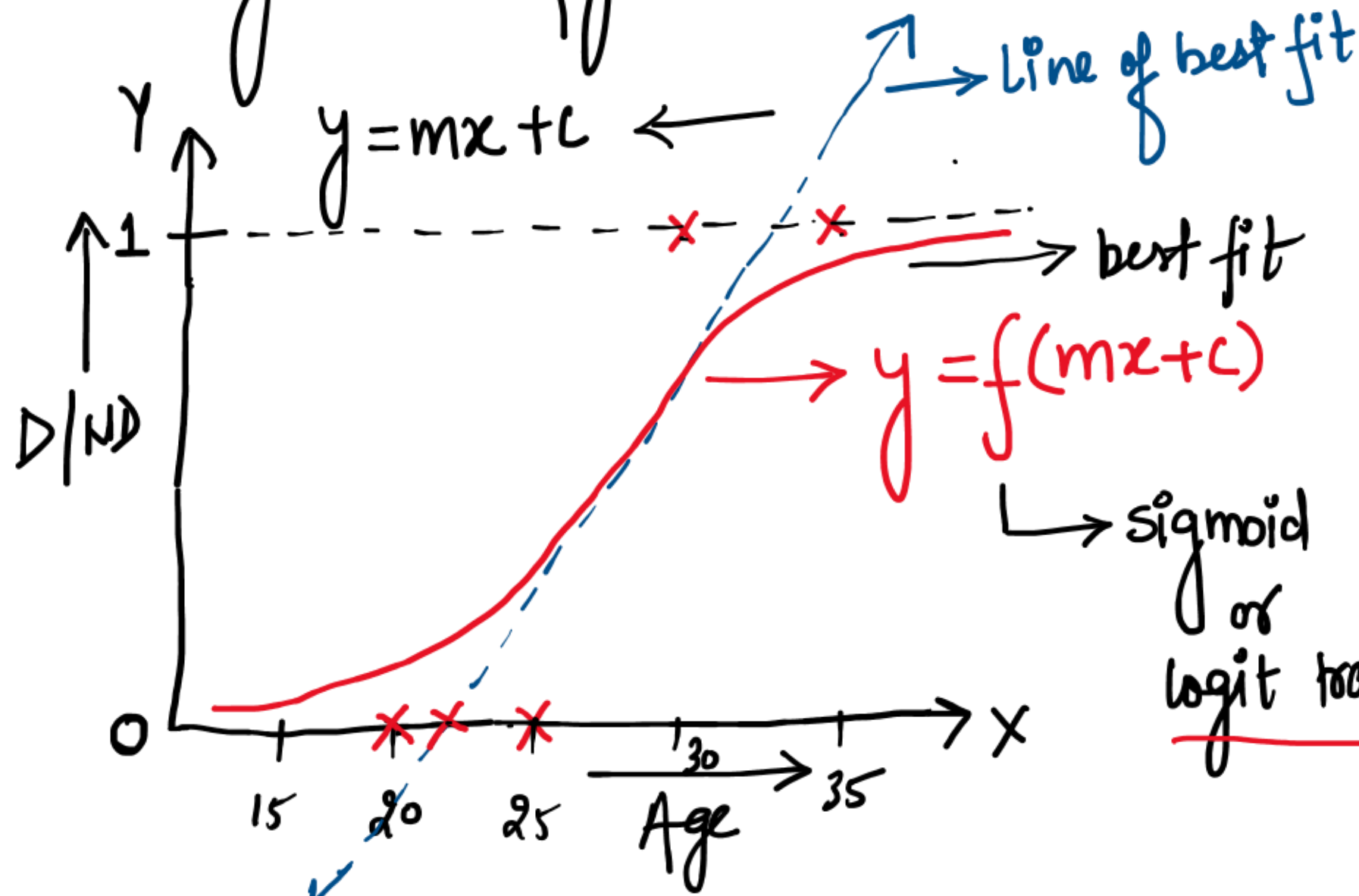
- house price
- demand of bikes
- Stock price
- Salary etc

Regression  
(Numerical)

- C/NC
- Y/ND
- F/NF
- C/NC
- D/ND

Classification  
↓  
[Logistic Regression] (Categorical)

# → Logistic Regression :- (Intuition)



Age  $\frac{D \approx 1}{ND = 0}$

| X  | Y |
|----|---|
| 20 | 0 |
| 30 | 1 |
| 22 | 0 |
| 25 | 0 |
| 35 | 1 |

→ Regression:-

$$y \in (-\infty, \infty)$$

→ Classificat<sup>n</sup>:-

↓  
probability

$$y \in [0, 1]$$

→ logistic Regression = logit  $f^n$  applied on top of Regression

→ Sigmoid transform<sup>n</sup>:- Ensure output  $\in [0,1]$

Mathematically,

$$f(z) = \frac{1}{1 + e^{-z}}$$

$$z \rightarrow \infty \rightarrow f(z) \rightarrow 1$$

$\frac{1}{1 + e^{-\infty}} \rightarrow 0$

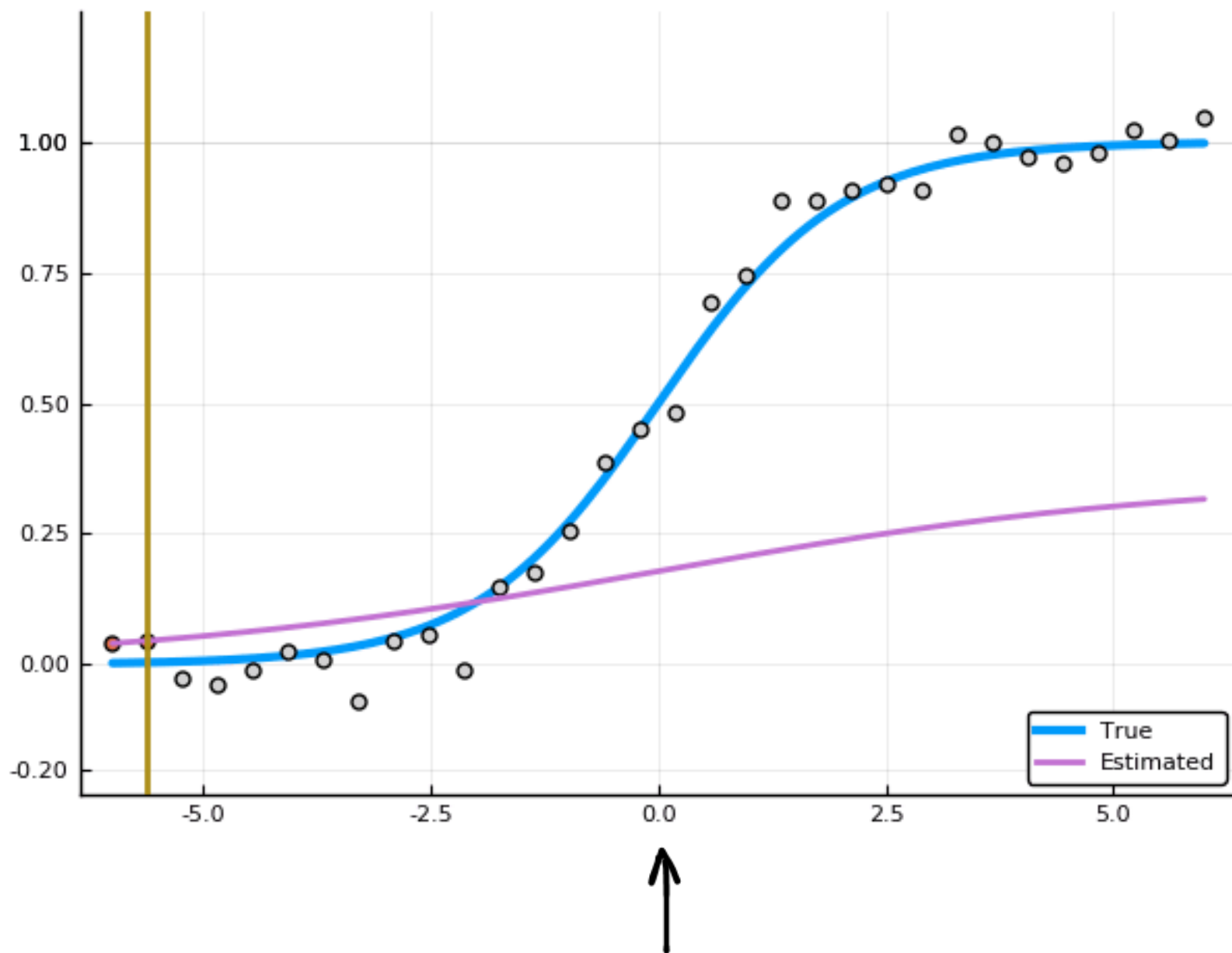
$$z \rightarrow -\infty \rightarrow f(z) \rightarrow 0$$

$\frac{1}{1 + e^{\infty}} = \frac{1}{V \cdot B \cdot N_0}$

$$y = f(mx + c) = \frac{1}{1 + e^{-(mx + c)}}$$

$$\frac{1}{1+e^0} = \frac{1}{1+1} = \frac{1}{2} = 0.5$$

$$\frac{1}{1+e^{-z}}$$



→ Odds:- Ratio of prob of happening ( $P(E)$ ) to the prob of not happening  $P(\bar{E})$

$$\text{odds} = \frac{P(E)}{P(\text{not } E)}$$

$\Rightarrow$

$$\boxed{\text{odds} = \frac{p}{1-p}}$$

✓

Example:- find the odds of getting 6 on die.  $\text{odds} = \frac{1/6}{5/6}$

$$\boxed{\text{odds} = 1/5}$$

→ log odds :-  $\log(\text{odds}) = \log\left(\frac{p}{1-p}\right)$  ✓✓

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→ logit  $f^n$  :-

$\log \text{odds} = \text{linear Equat}^n$

Assumpt<sup>n</sup>

$\left. \log\left(\frac{p}{1-p}\right) = mx + c \right\}$

---

$$\textcircled{1} \rightarrow \log\left(\frac{p}{1-p}\right) = mx+c$$

$$\textcircled{2} \rightarrow \frac{p}{1-p} = e^{mx+c}$$

$$\textcircled{3} \rightarrow p = (1-p) e^{mx+c}$$

$$\textcircled{4} \rightarrow p = e^{mx+c} - p e^{mx+c}$$

$$\textcircled{5} \rightarrow p + p e^{mx+c} = e^{mx+c}$$

$$\textcircled{6} \rightarrow p(1 + e^{mx+c}) = e^{mx+c}$$

$$\textcircled{7} \rightarrow p \frac{(1 + e^{mx+c})}{e^{mx+c}} = 1$$

$$\textcircled{8} \rightarrow p \left( \frac{1}{e^{mx+c}} + \frac{\cancel{e^{mx+c}}}{\cancel{e^{mx+c}}} \right) = 1$$

$$\textcircled{9} \rightarrow p \times (e^{-(mx+c)} + 1) = 1$$

$$\textcircled{10} \rightarrow \boxed{p = \frac{1}{1 + e^{-(mx+c)}}} \checkmark$$



$$y = \frac{1}{1 + e^{-(mx+c)}}$$

$$p \times t = 1$$

$$p = 1/t$$

$$y = \frac{1}{1 + e^{-(m_1 x_1 + m_2 x_2 + \dots + c)}}$$

$$y = \underline{\underline{mx+c}}$$

$$\log\left(\frac{p}{1-p}\right) = mx+c$$

$$p = m \cancel{x} + c$$

→ Interpreting of coeff of logistic Regression :-  $m = 0.7$   
 $C = 2$

$$\textcircled{y} = 10x + 3$$

$x \uparrow$   
1

$y \uparrow$   
10

$$\log\left(\frac{p}{1-p}\right) = 0.7 \times \textcircled{x_1} - 2$$

loan prob

job stability



→ A stable job increases the log odds of default by 0.7

$$\rightarrow y = \frac{1}{1 + e^{-(0.7x_1 - 2)}}$$

$$y_0 = \frac{1}{1 + e^{-2}}$$

$$y_0 = 0.8808$$

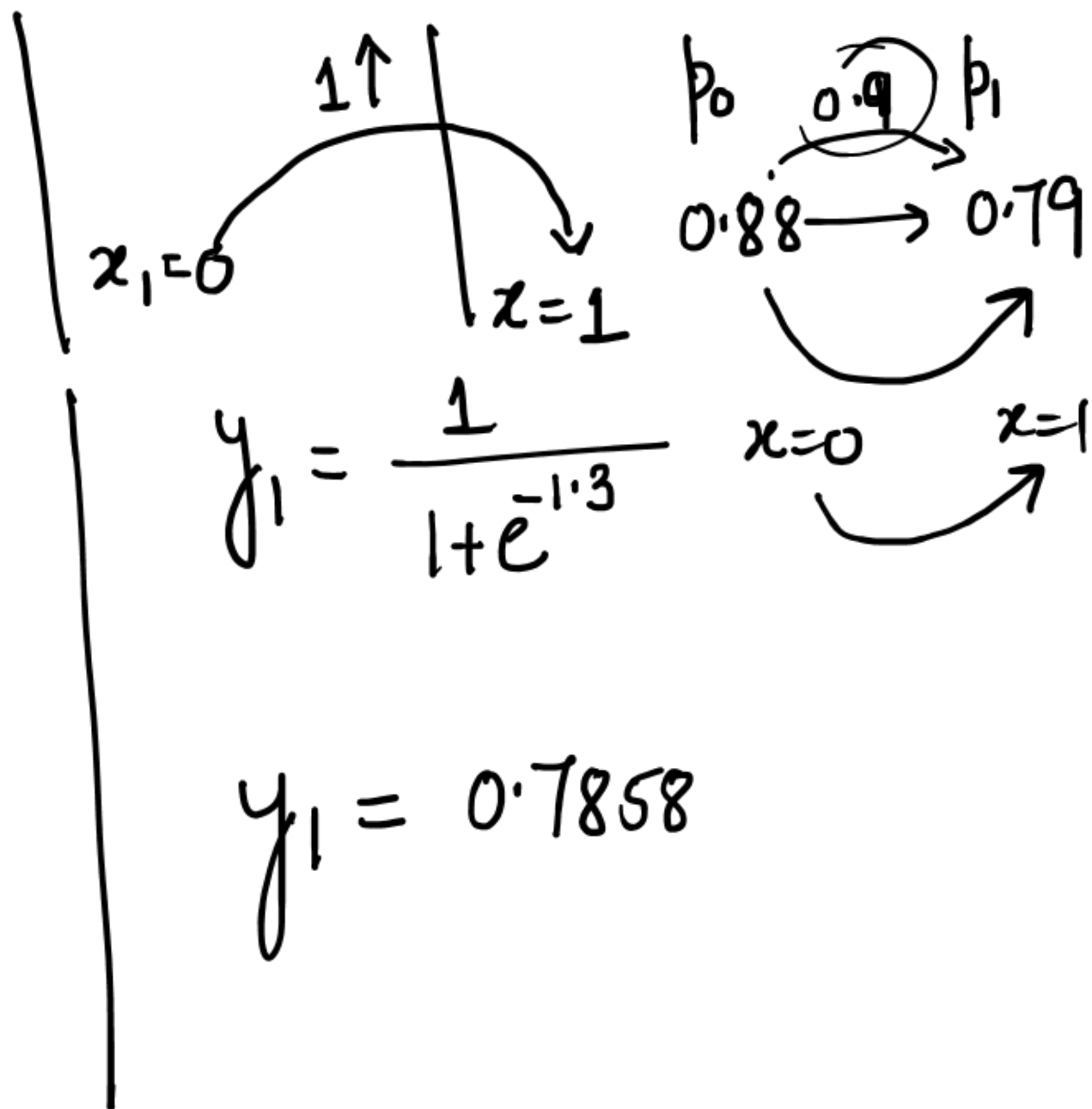


Diagram illustrating a neural network layer with two inputs,  $x_1 = 0$  and  $p_0 = 0.9$ , and two output nodes.

The first input node is labeled  $x_1 = 0$ . The second input node is labeled  $p_0 = 0.9$ . The output of the first node is  $y_1 = 0.7858$ . The output of the second node is  $0.79$ .

The diagram shows the following connections and values:

- Input  $x_1 = 0$  is connected to the input of the first sigmoid function.
- Input  $p_0 = 0.9$  is connected to the input of the second sigmoid function.
- The output of the first sigmoid function is  $y_1 = 0.7858$ .
- The output of the second sigmoid function is  $0.79$ .

$$\log\left(\frac{p}{1-p}\right) = 0.7x_1 - 2$$

$x_1 = 0$

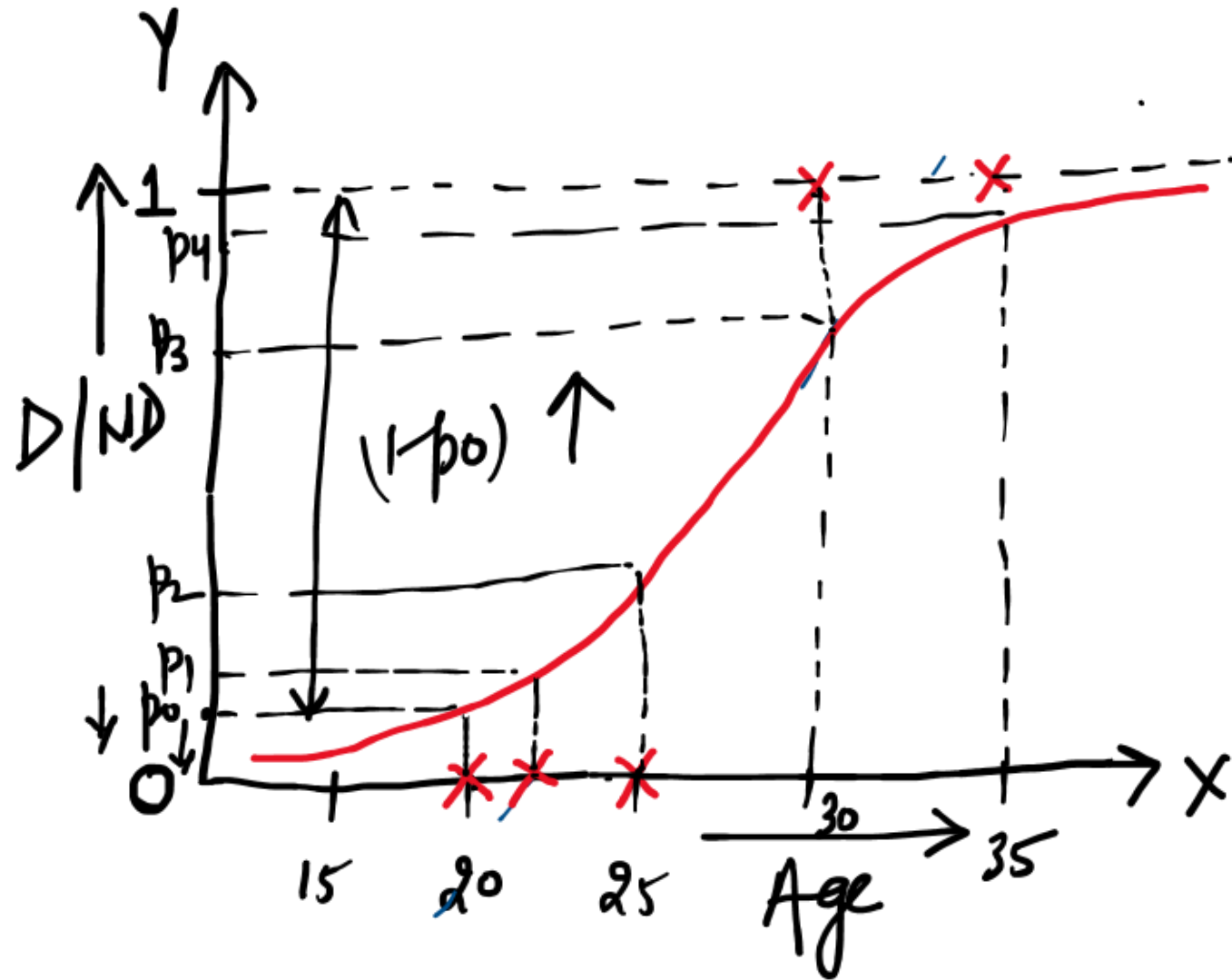
$$\log\left(\frac{p_0}{1-p_0}\right) = -2$$

$x_1 = 1$

$$\log\left(\frac{p_1}{1-p_1}\right) = -1.3$$

$$\log\left(\frac{p_1}{1-p_1}\right) - \log\left(\frac{p_0}{1-p_0}\right) = 0.7$$

→ Cost f<sup>n</sup> :- Maximum likelihood Estimat<sup>n</sup> (MLE)



Age  $\frac{D=1}{ND=0}$

| X  | Y |
|----|---|
| 20 | 0 |
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$p_0 \downarrow$   
 $p_1 \uparrow$   
 $p_2 \downarrow$   
 $p_3 \downarrow$   
 $p_4 \uparrow$

$\rightarrow L \downarrow$

$p_{y=0} \rightarrow \text{Min}$

$p_{y=1} \rightarrow \text{Max} \uparrow$

①

$\Sigma$

$\pi$

$\downarrow$

$(1-p)_{y=0} \rightarrow \text{Max} \uparrow$

$N$

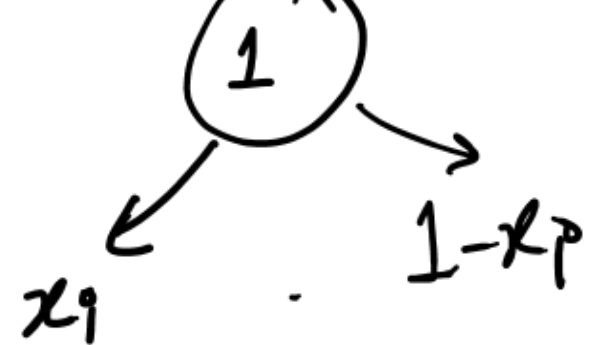
②

$\downarrow$

LE

Max  $L = \underbrace{(1-p_0)(1-p_1)(1-p_2) \dots}_{y=0} \times \underbrace{p_{n2} \times p_{n1} \times p_n}_{y=1}$  ③

easy best mle  $\rightarrow L = \prod_{i=1}^N p^{x_i} (1-p)^{1-x_i}$



$$\log a^b = b \log a$$

$$\begin{aligned} \log L &= \sum \log p^{x_i} (1-p)^{1-x_i} \\ &= \sum \log p^{x_i} + \log (1-p)^{1-x_i} \end{aligned}$$

$\log L = \sum x_i \log p + (1-x_i) \log (1-p)$

 ✓

→ Evaluation Metrics:-

$\left\{ \begin{array}{l} 1 \rightarrow + \\ 0 \rightarrow - \end{array} \right\}$

→ Confusion Matrix:- 2x2 matrix

|        | Predicted |     |    |
|--------|-----------|-----|----|
|        | T/F       | P/N |    |
| Actual | +         | TP  | FN |
|        | -         | FP  | TN |

→ T → A = P

→ F → A ≠ P

→ P = Predicted class +ve

→ N = Predicted class -ve

↑ TP → True Positives (+ → +)

↑ TN → True Negative (- → -)

FN → False Negative (+ → -) ↓

FP → False Positive (- → +) ↓



|     | Actual | Predicted |    |
|-----|--------|-----------|----|
| 1.  | 1      | 0         | FN |
| 2.  | 0      | 1         | FP |
| 3.  | 1      | 1         | TP |
| 4.  | 0      | 0         | TN |
| 5.  | 0      | 1         | FP |
| 6.  | 1      | 0         | FN |
| 7.  | 1      | 1         | TP |
| 8.  | 1      | 1         | TP |
| 9.  | 0      | 0         | TN |
| 10. | 0      | 0         | TN |

$N=10$

A

|   | + | - |     |
|---|---|---|-----|
| + | 3 | 2 | = 5 |
| - | 2 | 3 |     |

5

Confusion matrix

$$TN = 3$$

$$TP = 3$$

$$FP = 2$$

$$FN = 2$$

$$\underline{\underline{\frac{3}{5} = 60\%}}$$

→ Accuracy: How many datapoints model predicted accurately out of total datapoints.

$$A = \frac{TP + TN}{TP + TN + FP + FN}$$

= 60%

↔ Recall / Sensitivity / True +ve Rate :- (TPR)

How many datapoints model able to recall  
as +ve out of total actual +ve.

$$R = \frac{TP}{TP + FN} = 60\%$$

→ Specificity:

How many datapoints model able to recall  
as -ve out of total actual -ve.

$$S = \frac{TN}{TN + FP}$$

→ Precision:- How many datapoints model to precisely predict true out of predicted true.

$$P = \frac{TP}{TP + FP} = 60\%$$

→ F1-Score:- Combination of Precision & Recall

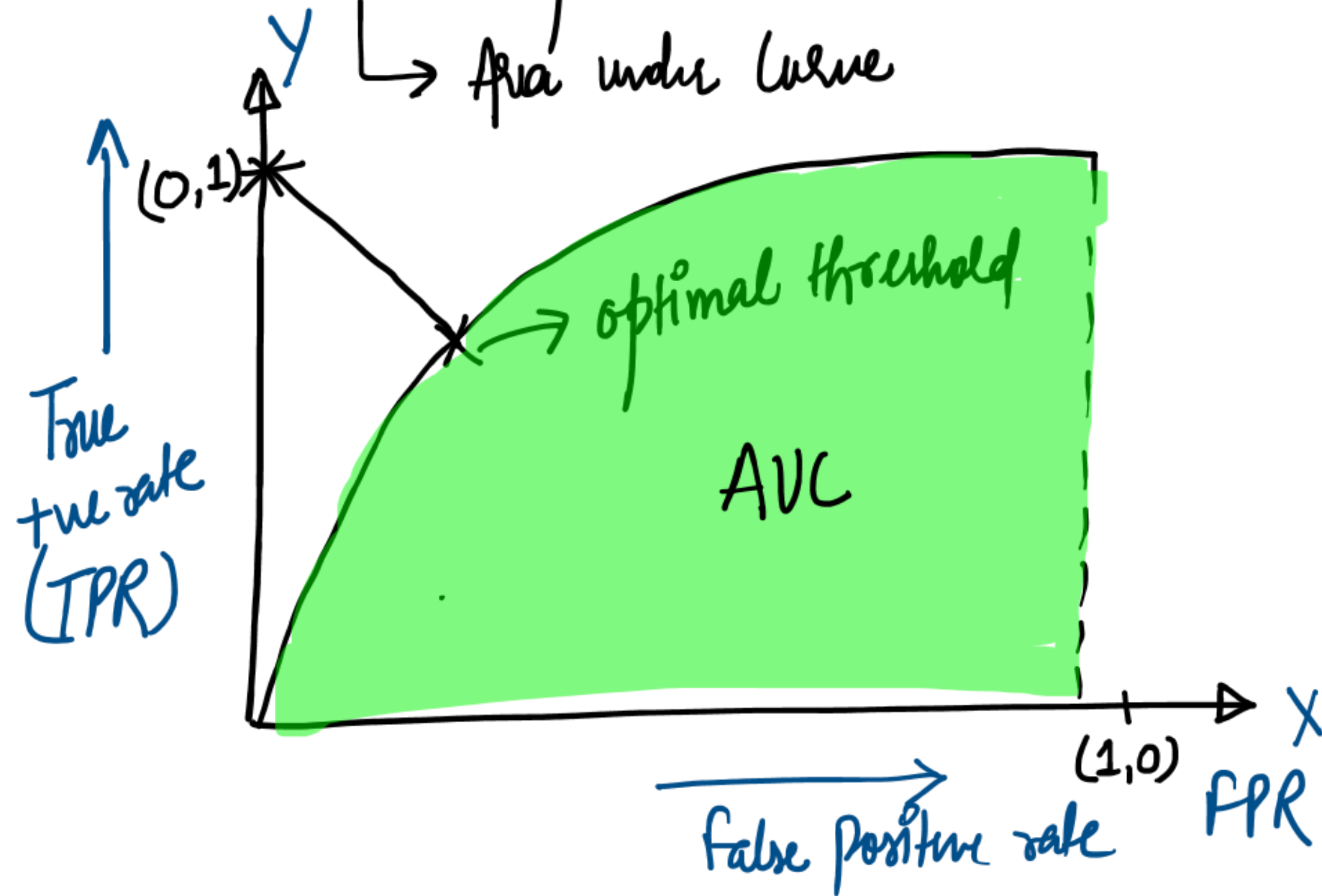
\* Harmonic mean of P & R.

$$\left( \frac{\frac{1}{P} + \frac{1}{R}}{2} \right)^{-1}$$
$$= \frac{P+R}{2PR}$$

✓ ↑

$$F_1 = \frac{2 \times P \times R}{P+R} \rightarrow [0-1]$$

→ ROC-AUC :- (Optimal threshold)



By default → 0.5

Receiving  
operating  
Characteristic  
Curve

0.1